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REVIEW PAPERS

Engineering interconnected cyber-physical-human systems for IoT applications: a review

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Abstract
The Internet of Things (IoT) is gaining popularity and usage in almost every field of human endeavor, enabling devices, objects or things, and computing gadgets to communicate using networks. This has recently introduced the concept of cyber-physical human systems, which is gaining momentum and has become an emerging area of research in IoT. A thorough understanding of the concept of cyber-physical human systems and exploring its length and breadth in enabling IoT guides researchers in assessing their bi-directional impact. This paper evaluated and analyzed approximately 170 articles (100 journals, 50 conference papers, and 20 reports) related to the importance and limitations of cyber-physical human systems in the IoT ecosystem. In addition, the study incorporated data from 50 IoT devices of different types and information on their applications. Additionally, a semi-structured interview has been conducted with 10 experts in IoT and engineering cyber-physical human systems, representing diverse industries such as healthcare, transportation, and manufacturing. The interviews aimed to gather insights on the state of IoT and engineering cyber-physical human systems, their applications, and future trends. It aims to underline the unrealized potential of technology and to bring forth difficulties and potential areas for engineering cyber-human physical systems in the IoT ecosystem. The study concludes by discussing essential findings and then by evaluating previous research and its limitations and suggestions for future work regarding IoT ecosystem-based engineering of cyber-physical human systems.

Keywords Cloud platform · Cyber-physical human systems · Machine learning · IoT · Security

1 Introduction

There hasn't been any other technology as innovative as IoT that has enjoyed the recent technological craze. According to estimates, the amount of IoT connections is anticipated to surpass 75 billion marks within the next few years (Gubbi et al. 2013; Cisco 2019). IoT's advent has diversified its application in new areas like education, healthcare, manufacturing, transportation, agriculture, and smart cities. The International Telecommunications Union authority has predicted that by 2030, there will be more than 25.4 billion IoT devices and connections globally (Cisco 2021). Many industries, such as healthcare, agriculture, transportation, and manufacturing, have started IoT development and are undergoing profound changes. Engineering Cyber Physical Human Systems (ECPHS) is a new interdisciplinary branch that combines Physical, cyber, and human systems to enhance the functioning and effectiveness of a complex system. ECPHS leverages IoT capabilities to address multifaceted challenges in smart cities, healthcare, and education, among other sectors. A defining characteristic of ECPHS is

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its ability to create intelligent, adaptive systems that autonomously learn and optimize performance, efficiency, and safety. By incorporating real-time data analytics and AI-driven decision-making, ECPHS enhances human-machine collaboration, making it a key enabler of next-generation IoT applications.

Despite the increasing adoption of IoT and the growing interest in ECPHS, existing research lacks a comprehensive evaluation of both academic and industry-driven aspects of ECPHS within the IoT ecosystem. On the other side, previous studies have summarized the development and benefits of ECPHS, but few have conducted an in-depth analysis of key research themes, core applications across various domains, and the challenges associated with implementing cyber-physical human systems in real-world IoT environments (Bello et al. 2024; Hakiri et al. 2024; Sicari et al. 2015).

This study explores recent IoT innovations and their impact on ECPHS by examining their evolution, significance, and growing role in various industries. A key objective is to provide a comprehensive overview of how IoT has shaped the development of ECPHS, highlighting its importance, benefits, and inherent limitations in IoT-driven environments. Additionally, the study assesses existing research themes and methodologies, identifying the approaches used to advance ECPHS in academic and industrial contexts. By analysing these trends, the study aims to uncover gaps in current knowledge and challenges in implementation. Furthermore, the research outlines key findings that contribute to a deeper understanding of ECPHS while identifying potential future directions for further innovation and improvement.

This study contributes to the field of ECPHS and IoT by:

- Conducting a systematic review of 170 research articles (100 journal papers, 50 conference papers, and 20 reports).
- Examining 50 IoT devices to understand their application in cyber-physical human systems.
- Gathering expert insights through semi-structured interviews with 10 healthcare, transportation, and manufacturing professionals.
- Developing an ECPHS framework that defines its key components and interactions.
- Providing recommendations for future research and industry adoption.

The remainder of this paper is structured as follows. The paper begins with Sect. 2 to introduce basic information about IoT alongside ECPHS alongside existing research publications which build foundation knowledge about the subject matter. The proposed ECPHS framework is introduced in Sect. 3 with descriptions of its elements and design concepts. Data collection and analysis methods implemented during the study form the basis of Sect. 4. Results with important research findings appear in Sect. 5 of the paper. This section analyzes present and potential difficulties alongside research possibilities within the field while providing evaluations for present and future innovations. The final section of this study presents concluding remarks about essential findings and suggests future research directions.

2 Background and related work

2.1 Overview of engineering cyber-physical human systems (ECPHS)

Located between automated engineering systems and human operators lies the realm of integrated ECPHS. ECPHS centers on creating intelligent robots, control devices, and corresponding algorithms designed for human-to-machine interaction for effective problem-solving. ECPHS is an under-researched area which includes the application of cybernetic, human, and physical systems (Bello et al. 2024; Dave et al. 2024; Cederbladh et al. 2024). Much of the research in ECPHS is dedicated to designing systems that support user interaction with complex systems and automating systems activities to enhance functionality (Oliveira et al. 2024).

2.2 Key features of ECPHS

Cyber-physical systems serve to supervise and administer physical processes efficiently and reliably. The confluence between the physics and cyber components domains is physically networked within a network. They are mainly designed to interact with physical components and control them using actuators and sensors in a feedback loop. They are systems that combine sensors, actuators, and computer systems to carry out a specific task. One of the critical features of Cyber's physical system is the real-time data processing capabilities as they usually interact with the physical systems or world as well as monitor the environment. They are shown to exhibit high reliability and safety, as errors or bugs can have severe consequences. A cyber-physical system monitors the physical system while creating a virtual copy and developing decisions with the networked ecosystem reliably and efficiently. Scalability is another critical feature in the discussion of cyber-physical human systems, given the humongous amount of data being generated and needs to be processed to get insight from them. Table 1 presents an organized structure demonstrating how physical,

Table 1 Framework for engineering cyber-physical human systems

Physical system	Cyber system	Human system
Smart grid	SCADA system	Control center
Smart building	HVAC control system	Occupants
Smart transportation	Traffic management system	Drivers, Passengers
Smart healthcare	Electronic health record system	Clinicians, Patients

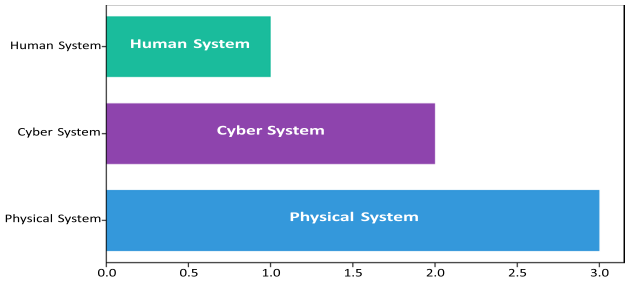


Fig. 1 Engineering cyber-physical human system frameworks

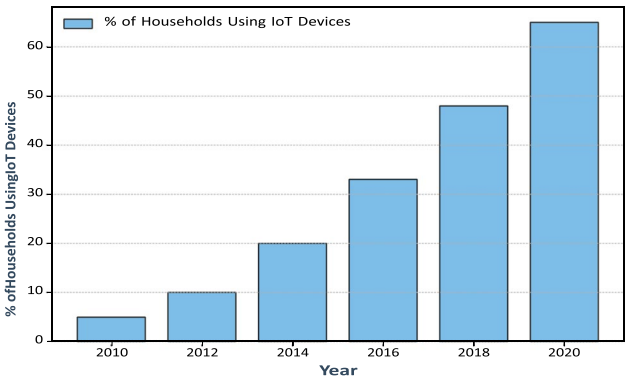


Fig. 2 Engineering cyber-physical human systems framework

Table 2 Engineering applications and the impact of cyber-physical human systems

Domain	Applications	Impact
Healthcare (Khater et al. 2024)	Remote patient monitoring, Smart hospital systems	Disease prevention, improved patient outcomes, cost savings
Transportation (Li and Jin 2024)	Smart traffic management, Connected vehicles	Reduced traffic congestion, improved safety, cost savings
Manufacturing (Wang et al. 2022)	Smart manufacturing systems, Industrial IoT	Improved efficiency, reduced downtime, and cost savings

cyber, and human systems relate to each other in ECPHS applications.

The engineering framework for Cyber-Physical Human Systems (CPHS) presented in Fig. 1 shows how physical elements connect with cyber and human aspects. The physical domain contains real-world sensors and actuators that enable the acquisition of data and the execution of actions. Computational models combined with artificial intelligence and communication networks enable the cyber system to process the provided data for optimal decision control. People from the human system must engage with both the physical and cyber layers to participate in decision-making and related adaptations. The unified framework represents an environment that supports quick data sharing, enabling intelligent automation and operational enhancement for smart manufacturing, healthcare, and autonomous systems applications.

2.3 IoT applications across industries

Through IoT technology, health providers can establish connected device networks for collecting and exchanging data, which enables ECPHS operations. They are self-operating systems that function with little human involvement because they use sensors and analytical tools for process optimization. Different sectors use IoT applications to improve operational efficiency, make better decisions, and reduce operational costs.

2.3.1 IoT adoption rate over time

The utilization of IoT devices has skyrocketed during the last ten years, according to Fig. 2. Data indicates that household use of IoT devices (HUIoTD) increased dramatically from 5% in 2010 to 70% during the next ten years between 2010 and 2020. The quick rise in IoT adoption rests on improved wireless connections, inexpensive smart devices, and the expanding need for automatic systems in everyday life. The growing integration of IoT devices became widespread mainly because of smart home solutions offering security systems, voice assistants, and energy management capabilities. Modern IoT solutions gained consumer appeal through improvements in artificial intelligence and cloud computing systems, making these solutions more accessible to customers. The ongoing trend indicates that IoT adoption will escalate to promote smarter living and better-connected environments.

2.3.2 Impact of IoT-enabled ecphs across domains

Table 2 provides detailed information regarding ECPHS applications alongside their effects in different domains. ECPHS technologies have transformed healthcare patient care through innovations, including remote patient monitoring and smart hospital systems in the medical field. The improved technology helps doctors find diseases early while providing active health tracking and analytic predictions, leading to better patient success rates, more efficient medical service delivery and major financial benefits.

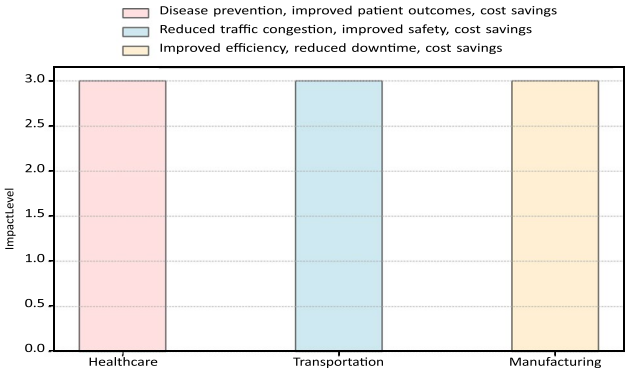


Fig. 3 Impact of engineering cyber-physical human systems across domains

Table 3 Key challenges and limitations of IoT implementation

Challenge/Limitation	Description
Security	Lack of security standards, vulnerability to cyber threats
Privacy	Risk of data breaches, unauthorized data collection
Interoperability	Compatibility issues among diverse IoT platforms
Scalability	Managing the vast amount of data generated by IoT devices
Cost	High infrastructure and maintenance costs

Artificial intelligence and IoT allow medical devices to interact with healthcare professionals through connected systems, enhancing diagnosis accuracy and patient-specific care plans. Wearable health devices combined with telemedicine services have improved healthcare accessibility for patients while increasing their involvement with healthcare services, cutting down hospital readmissions and enhancing disease management.

Smart traffic management systems and connected vehicle technologies significantly improve transportation operations through reduced traffic delays, advanced road protections, and reduced transport expenses. Reliable time-based data analysis combined with AI-assisted infrastructure communication and automated decision strategies helps transportation systems reach maximum efficiency and benefit from accident prevention. Besides improving driving efficiency and fuel economy, integrating autonomous and semi-autonomous vehicle technologies has substantially decreased human driving mistakes. The manufacturing sector experiences improved production processes and maintenance ability and decreased operational interruptions using smart manufacturing systems and Industrial IoT platforms. Manufacturing processes enhance supply chain management and productivity and achieve cost reductions through connected sensor deployment, automated AI systems, and real-time data analytics. A visual depiction presented in Fig. 3 demonstrates how ECPHS technologies continue advancing numerous businesses.

2.4 Challenges and limitations of IoT in ECPHS

Organizations face several barriers while integrating IoT technology, as shown in Table 3. Five broad challenges and restrictions have been identified: security, privacy, interoperability, scalability, and expense. The security challenge concerns the absence of minimum controls or standards to defend IoT devices from cyber-attacks. The privacy challenge concerns the potential collection and retention of personal data, which can be exposed through data breaches. The interoperability challenge denotes the gaps in compatibility among various IoT devices and platforms. The limitations on scalability refer to the control of vast quantities of data resulting from IoT devices. Lastly, the financial burden challenge encompasses the expense of the IoT devices and infrastructure as well as continual maintenance expenditure. These challenges must be managed effectively if organizations hope to gain the benefits of IoT technology.

2.5 Existing work on IoT-ECPHS integration

Several studies have researched the merger of IoT with ECPHS and other technologies. These studies forecast the possible benefits of merging IoT and ECPHS, such as improved resource conservation, productivity, and better decision-making (Mu and Antwi-Afari 2024). However, this merger poses problems, such as the need for sophisticated sensors and advanced-level data processing (Tariq et al. 2024). The reviewed studies have focused on possibilities of ECPHS integration with IoT which include enhanced energy management, increased productivity, and better decision-making capabilities (Mu and Antwi-Afari 2024). Moreover, such integration has challenges, such as the need for novel sensors and advanced data analytics technology (Tariq et al. 2024). The iterative and comparative review of literature on engineering cyber-physical human systems in the IoT ecosystem has been summarized in Table 4.

The exploration of IoT-ECPHS integration has produced meaningful results; however, many fundamental gaps exist regarding practical implementation, security issues, and interoperability problems. Theoretical frameworks proposed for IoT-ECPHS integration fail to demonstrate evidence through testing in dynamic operational conditions, even though these assessments determine their effectiveness in changing environments. Integrating various IoT devices into a cyber-physical system is faced with the classical issues of merging disparate IoT devices, cybersecurity risks, and privacy issues. Future research should focus on creating stronger security boundaries, semantic interoperability, and

Table 4 Comparative analysis of existing studies on IoT-ECPHS integration

References	Aim and objectives	Methodology	Limitation (s)	Strength (s)
Koch (2024)	Proposes a framework for modelling cyber-physical human systems in the IoT ecosystem.	Literature review and model development.	The framework has not been tested in a real-world scenario.	The framework can be used to design and evaluate cyber-physical human systems in the IoT ecosystem.
Piardi et al. (2024)	Investigates the integration of machine learning and IoT in engineering cyber-physical human systems.	Literature review and case studies.	The case studies are limited to healthcare and industrial applications.	The study provides insights into the potential of machine learning and IoT in improving engineering cyber-physical human systems.
Alsamhi et al. (2024)	Develops a human-centric framework for designing cyber-physical systems in smart cities.	Literature review and framework development.	The framework has not been tested in a real-world scenario.	The framework underscores the significance of human factors in the design of cyber-physical systems within smart cities.
Aljabri et al. (2024)	Explores the use of blockchain technology in cyber-physical systems.	Literature review and case studies.	The study focuses on blockchain's impact on cyber-physical systems' security and privacy.	The study provides insights into the potential of blockchain technology in improving the security and privacy of cyber-physical systems.
Mendia et al. (2024)	Develops a framework for optimising energy consumption in cyber-physical systems.	Literature review and model development.	The framework has not been tested in a real-world scenario.	The framework can be used to optimise energy consumption in cyber-physical systems and reduce their environmental impact.
Thakur and Sehgal (2024)	Investigates the use of edge computing in cyber-physical systems.	Literature review and case studies.	The framework emphasises the importance of considering human factors in the design of cyber-physical systems for smart cities.	The research offers valuable insights into how edge computing can enhance the performance and efficiency of cyber-physical systems.
Kanso et al. (2024)	Develop a framework for designing context-aware cyber-physical systems.	Literature review and framework development.	The framework has not been tested in a real-world scenario.	The framework can be used to design context-aware cyber-physical systems that can adapt to changing environments.
Hasan et al. (2024)	Explores the use of deep learning in cyber-physical systems.	Literature review and case studies.	The research centres on investigating the influence of deep learning on the performance and efficiency of cyber-physical systems.	The research offers valuable insights into how deep learning can enhance the performance and efficiency of cyber-physical systems.
Buyya et al. (2024)	Develop a framework for energy-efficient cyber-physical systems.	Literature review and framework development.	The framework has not been tested in a real-world scenario.	The framework can be used to design energy-efficient cyber-physical systems that can reduce their environmental impact.
Li et al. (2018)	Investigates the use of fog computing in cyber-physical systems.	Literature review and case studies.	The research centres on examining the influence of fog computing on the performance and efficiency of cyber-physical systems.	The research offers valuable insights into how fog computing can enhance the performance and efficiency of cyber-physical systems.

industrial-scale optimization of the system's infrastructure. Provide a better explanation of how emerging technologies, specifically artificial intelligence, blockchain technology, and edge computing, can be integrated to improve system efficiency, security, and adaptability. These technologies and proper integration allow for better adoption of IoT-ECPHS systems across various sectors and industries.

3 Proposed framework: ECPHS

For the development of intelligent, interconnected systems, engineers implement the integration of physical elements with cyber and human systems using the ECPHS framework. Besides containing and describing their networks, ECPHS contains all critical components and has a systematized framework for creating industrial and urban cyberspace implementations. This part offers particulars on two sides.

3.1 Key components of ECPHS

The three main parts of the ECPHS are the set of physical infrastructure, the human operators, and information technology. All the system components work collaboratively to achieve automation, decision-making, and real-time operations control for different types of exercises.

3.1.1 Physical systems

The entire system concept of ECPHS includes the infrastructure in different locations and systems that capture and process data. The distribution of electric energy is optimized by IoT sensors working with real-time monitoring and automated control systems, enabling maximum efficiency while preventing cutoffs and accepting renewable energy resources. SCADA systems are fundamental devices for studying the energy behavior of customers and predicting their demand within certain periods. In healthcare, IoT starts with medical devices that gather data from patients in real time through remote patient monitoring systems and electronic health records, resulting in automated alerts with predictive analytics, which allows for modifying treatment plans and speeding up the response time of patients. Smart systems integrated into vehicles and automatic toll collection devices allow real-time streaming of traffic management, enhancing orderly vehicle movement and improving public transport service. Combining IoT sensors and real-time data processing tools is essential for reducing congestion while promoting better mobility. Industrial IoT (IIoT) platforms within manufacturing systems help improve three main aspects: predictive maintenance and automated processes, and enhanced supply chain operations, which fuel optimized production processes and operational efficiency. These physical systems produce massive amounts of data that require connection to cyber and human systems to support decision-making and regulatory functions.

3.1.2 Cyber systems

ECPHS operate through cyber systems to achieve digital integration of data collection and processing and communication and automation functions. Such IoT networks allow devices to exchange data efficiently through protocols MQTT and CoAP which work well in resource-sensitive operating systems. Through cloud computing, users gain both flexible storage capabilities and high-speed computing resources that let them conduct big data analytics alongside predictive modelling AI-based automation operations and remote data access. The performance of edge nodes improves because edge computing operates data near its origins to execute real-time transformative analytics and critical on-site decisions. Product functions like healthcare and transportation need rapid responses, particularly this type of system infrastructure. System intelligence receives additional enhancement through AI-based processing because machine learning models analyze sensor data to detect anomalies simultaneously with failure prediction for performance optimization, automated system improvements, and improved collaboration between humans and machines. The cyber systems collectively support fast real-time data analysis and maintain secure communication links between physical hardware and human control systems and operational efficiency.

3.1.3 Human systems

ECPHS requires human systems to properly align technology and human needs with safety standards and decision-making processes. A system of control centers, both centralized and distributed, operates ECPHS by tracking performance together with the identification of abnormalities and execution of both automated and manual commands. Controllers use dashboards, real-time alerts, and analytics tools to interface with cyber systems. End users, comprising consumers and patients, industrial workers, and drivers, link with ECPHS through automated services, interfaces, and applications that feature user-friendly IoT applications for smooth connection to smart environments. Policymakers, industrial leaders, and urban planners can better allocate resources for regulatory needs using ECPHS analytical data, and AI analytics improve operational decision-making. Decision-makers use these systems to improve safety operations and standards through enhanced resource optimization. Ergonomic and safety-compliant features of ECPHS are due to the successful implementation of the system's human components. Collaboration between humans and machines is aided by intelligent assistive robotic automation and wearable devices, which enable seamless collaboration between cyphers and physical systems to aid in efficient work.

3.2 Interaction between physical, cyber, and human systems

Specialists responsible for ECPHS use the bilateral interaction between human, cyber and physical systems for automation and adaptive decision-making. The boundary of the environment, with the aid of sensors, operates Informatics technology devices such as IoT gadgets, which enable these devices to relay physically perceived data information systematically so that analysis and decision automation can take place. Physically responsive automated control systems operate under the PF process, which includes changing

power grid distributions, adjusting traffic control systems, and turning on the emergency healthcare system. Engagements of humans and cyber systems are feasible if users access cyber systems via dashboards, mobile applications, and automated reporting interfaces. Similarly, strategists utilize performance optimization AI-driven analytics to assist them in making better decisions. The last type of interaction integrates human operators with mechanical devices that respond to voice commands and controllers or wearable technologies. The interconnected structure of these components allows for further seamless intercommunication among the elements and, thus, holistic, intelligent system development and increased system adaptability.

3.3 Proposed ECPHS framework

It is still of prime importance to engineering standardized ECPHS frameworks because they improve integrating disparate physical systems with cyber networks and human systems. The offered framework has a multilayered structure.

- Data Collection Layer: The first layer gets information from the Data Collection Layer, which comprises IoT devices, sensors, and embedded systems in the physical domains. Each system's components performs continuous conservative temperature evaluation along with humidity and motion measurements, pressure indicators, and system operational status to guarantee continuous data collection. Wireless sensor networks (WSNs) and RFID technology implementations make the data collection layer more efficient and accurate.
- Data Processing Layer: The data is processed in real-time using Edge computation nodes along with cloud platforms, which work on filtering and storing data. Since edge devices are located close to the analysis site, they help lower the response time and save network traffic before cloud solutions are used for bulk analyses. This layer incorporates data fusion strategies that combine information from several sources to provide comprehensive situational awareness about the system's functioning.
- Decision-Making Layer: Engineers apply modern parsers between machines and AI with control loops that maximize the operational effectiveness of the machine and system using real-time telemetry data. The predictive analytics interface combined with deep learning and reinforcement learning enables the detection of patterns and anomalies while performing failure predictions. Human operators receive better decision-making support through decision-support systems, enhancing the reliability and intelligence of cyber-physical human systems.
- Action and Feedback Layer: Real-time conditions drive the behaviour adjustments of automated responses and alarms monitored through human interventions. The actuation systems, robotic components, and automatic control protocols perform corrective measures using continuous feedback control systems for optimization purposes. Manual human involvement through HITL (Human-in-the-loop) mechanisms provides the required balance by letting personnel step in if necessary. This layer utilizes security measures that protect operation systems from unauthorized access and system vulnerabilities to guarantee reliability and security during operations.

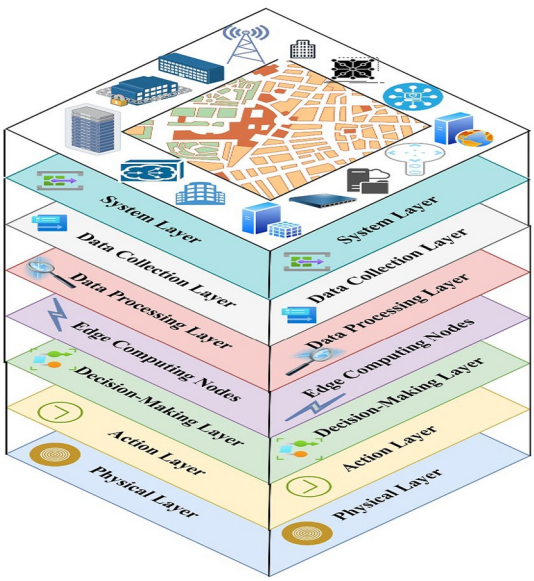


Fig. 4 Proposed ECPHS framework

The ECPHS framework shows physical, cyber, and human system interactions through Fig. 4. The model demonstrates how processed data moves across connected layers to create efficient system responses. Real-time operation of smart sensors, actuators, and IoT devices forms the basis of the Physical Layer, which continuously monitors and collects data from environmental conditions. The initial data goes through the Data Processing Layer after being transmitted from the Physical Layer to specific edge nodes and cloud-based infrastructure. AI models running machine learning algorithms inside the Decision-Making Layer examine processed data to find patterns while making predictions that lead to operation improvements. Between these layers exists a feedback mechanism which establishes a real-time interactive process that improves the decision-making process for smart grids and solutions for healthcare manufacturing and transportation.

The ECPHS framework structure presents itself in Table 5, which provides step-by-step details about the main

layers, components, and operational functions. Environmental data becomes available for intelligent system responses through the Data Collection Layer, which contains sensors and IoT devices that gather instant information. Data Processing Layer consists of edge computing nodes with cloud infrastructure that acts as a system to process substantial data collection efficiently and perform data filtering followed by storage for future analysis. The Decision-Making Layer uses AI-powered machine learning algorithms to translate processed data into actionable insights, predictions, and automated decisions. The hierarchical design enables smooth data flow and improved operation efficiency and adaptive capabilities, making it a reliable structure for cyber-physical-human framework implementation throughout various industrial and societal programs.

4 Key insights

The research methodology, together with data collection approaches and analysis methods for studying ECPHS integration with IoT, is presented here. A qualitative research approach includes literature review analysis, physical device assessment, and expert interview information. Thematic analysis techniques evaluated the obtained data to uncover vital patterns alongside difficulties and research prospects related to IoT-based ECPHS.

4.1 Research approach

Researchers can study actual IoT system deployments alongside their obstacles and technological advancements through this method. The study examines modern IoT developments to deliver critical information about how cyber-physical human systems integrate across different industrial sectors and social domains. Additionally, the case study method facilitates the identification of emerging trends, technological gaps, and potential areas for innovation. The research outcomes yield information which advances the creation of more powerful and flexible cyber-physical human systems that scale to different applications.

Table 5 ECPHS framework layers and their functions

Layer	Components	Function
Data collection layer	Sensors, IoT devices	Captures real-time data from physical environments.
Data processing layer	Edge nodes, Cloud infrastructure	Processes, stores, and filters raw data.
Decision-making layer	AI Models, Machine learning	Analyses data, predicts outcomes and makes decisions.
Action and feedback layer	Automated responses, Alerts, Human Input	Implements decisions and adapts system operations.

4.1.1 Research questions

Several research inquiries have been developed to match the goals of this research while supporting the selection of appropriate literature during the review stage. The study analyses contemporary ECPHS practices in IoT ecosystems to identify research barriers preventing their general use. These research questions create an organized foundation to investigate existing studies for important knowledge extraction. The investigation examines the most recent ECPHS integration approaches in IoT and looks at their applications across specific domains plus factors affecting operational efficiency and scalability limits. Through this study, researchers identify key application areas where ECPHS requires additional enhancement for innovation. The questions presented below guide the selection of suitable research publications about the subject to maintain an unbiased and complete literature review which assesses current trends and future potential in the field.

- What are the modern research results concerning IoT-related integration of engineering systems with cyber-physical human components?
- Which part of the IoT infrastructure does the cyber-physical engineering human system operate within?
- The study area faces what limitations?
- The future development of IoT depends on which aspects can be improved by engineering cyber-physical human systems.

The study works to resolve key questions that link conceptual gaps while offering important information to researchers and industry experts to help forecast the development path of ECPHS within IoT platforms.

4.2 Data collection

The study used data collection techniques such as document analysis, observation, and semi-structured interviews to gather information on IoT and engineering cyber-physical human systems. The documents analyzed included academic journals, conference papers, reports, and online resources covering various topics related to IoT and engineering cyber-physical human systems, including their applications in healthcare, transportation, and manufacturing. The search for relevant documents is done using search engines like Google Scholar. This report summarizes the observations made at three manufacturing sites, two hospitals, and one transportation centre to identify new IoT

devices along with their use cases in various sectors. The observation is conducted on 50 devices, including sensors, smart cameras, and smart wearable devices. Remote Patient Monitoring, Traffic Management, and Predictive Maintenance for Manufacturing Devices are some of the use cases. The identification details of the hospitals and the other two facilities are used in the report to support the observation. To get a multi-perspective view, the sample included ten specialists in IoT and engineering of cyber-physical human systems who intersect with diverse fields - healthcare, manufacturing, and transportation. The participants are chosen based on their knowledge and skill set in the subject matter. The interviews were done face-to-face and remotely via video calls, each scheduled for 30 min. The main goal of the interviews is to get relevant information regarding the known state, use, and prospects of IoT and the engineering of cyber-physical human systems. Google Scholar and EBSCOhost are some of the databases used for document search. Between the years 2015 to 2024, 100 academic journals, 50 conference papers, and 20 reports focusing on IoT and engineering cyber-physical human systems have been obtained and analyzed. Out of 170 papers, we have included 33 of the best articles in this manuscript. The study collected, compiled, and presented application data from 50 different types of observed IoT devices in tables within the report.

Additionally, 10 industry experts have been interviewed using a semi-structured format. Among them, 5 represented the healthcare sector, four the transportation sector, and one the manufacturing sector. The primary objective of these interviews is to explore how existing IoT and engineering cyber-physical human systems are utilized and to predict future developments in the field. Each interview lasts approximately 30 min and is conducted through face-to-face meetings and video conferencing. Table 6 presents the demographic details of the study respondents regarding age, sex, educational background, and occupation.

Table 6 Demographic characteristics of participants

Demographic characteristic	Number of participants
Age	18–25: 15, 26–35: 25, 36–45: 20, 46–55: 10
Gender	Male: 45, Female: 25
Education	Bachelor's degree: 30, Master's degree: 35, PhD: 5
Profession	Software Engineer: 25, Data Scientist: 15, IT Consultant: 20, Researcher: 10, Other: 20

Table 7 IoT devices analyzed across sectors

Sector	IoT applications	Impact
Healthcare	Remote monitoring, smart hospital systems	Improved patient outcomes, cost savings
Transportation	Traffic management, connected vehicles	Reducedcongestion, improved safety
Manufacturing	Smart factories, predictive maintenance	Enhancedefficiency, reduced downtime

4.2.1 IoT Device analysis

To complement the literature review, an observational study is conducted on 50 IoT devices across multiple industries, focusing on real-world applications, operational limitations, and their impact on ECPHS efficiency. In healthcare, remote patient monitoring systems, smart hospital management, and wearable health trackers have been analyzed for their role in improving patient outcomes, enhancing real-time diagnostics, and reducing operational costs. In transportation, smart traffic management systems, connected vehicle infrastructure, and automated toll collection technologies have been assessed for their effectiveness in reducing congestion, improving road safety, and optimizing traffic flow. In manufacturing, IIoT solutions such as predictive maintenance, automation, and supply chain optimization are evaluated for their impact on efficiency, production reliability, and cost reduction. Data in Table 7 demonstrates that IoT applications enhance patient healthcare standards while saving costs, improving transportation safety through congestion reduction, and strengthening manufacturing operations by reducing downtime. The findings demonstrate the essential functions IoT-enabled ECPHS facilitates for sector automation and real-time operational effectiveness.

Multiple challenges exist during IoT system deployments despite their multiple advantages. Security vulnerabilities emerge from inadequate encryption practices and unauthorized access threats that endanger IoT networks' integrity. Multiple interoperability problems exist because vendors cannot connect their IoT devices seamlessly, thus creating obstacles to coordinated communication and operation between varied IoT devices. Data storage and processing requirements get out of hand with sensor-generated volumes because existing systems struggle to handle the growing scale, so advanced solutions are necessary. Device deployment, maintenance costs, and IoT system updates are major obstacles to broad implementation due to their high financial requirements. The analysis revealed essential practical barriers to ECPHS-IoT integration by showing the necessity to build stronger security systems, improve integration capabilities, and design flexible systems.

4.2.2 Expert interviews

The research conducted semi-structured interviews with ten domain specialists who addressed topics in IoT technology, cyber-physical systems, and industrial automation and operated within healthcare services, manufacturing sectors, and transportation domains. These specialists delivered

multi-domain insights concerning ECPHS implementation, related hurdles, and predicted tendencies through their expert knowledge and active project involvement in IoT-ECPHS development. The research interviews covered four main topics: the current pace of IoT adoption, difficulties with system expansion and security requirements, and human-machine interface improvement while discussing AI edge computing and blockchain in ECPHS applications and illustrating real-world organizational successes using IoT. Experts showed their interest in the rising IoT automation of real-time systems. Still, they emphasized how interoperability issues, cybersecurity risks, and clear human-machine interfaces remained significant concerns. The experts recognized AI alongside edge computing and blockchain as important tools which boost efficiency together with security capabilities and enable real-time execution. The experts analyzed are grouped into three sectors, with healthcare professionals comprising five participants, while the transportation sector had four experts alongside one professional from manufacturing. The expert panel included healthcare and transportation professionals who shared their knowledge on connected patient care systems and hospital automation initiatives, and the manufacturing expert revealed an understanding of predictive care, automated operations and optimization. The discussions provide essential industry expertise about how IoT-ECPHS evolves in essential industries (Table 8).

Voice-recorded interviews ran for 30 min and took place using both face-to-face meetings and video platforms. Because of its nature, the interviews collected in-depth responses, allowing researchers to uncover detailed perspectives regarding IoT-ECPHS developments and their implementation challenges and new technological possibilities. These expert insights transform into important knowledge points that explain industrial standards and clarify regulatory demands and the developing function of IoT within cyber-physical human systems.

4.3 Data analysis and evaluation

Expert analysis of IoT integration in ECPHS identified major findings through research studies, industrial applications, and professional expert assessments. The review of ECPHS frameworks highlighted security weaknesses and interoperability problems, and an IoT device assessment evaluated deployment practices, performance indicators, and implementation barriers regarding healthcare, transportation, and manufacturing. Security threats, scalability challenges, and emerging technologies such as AI blockchain and edge computing have been examined through expert interviews conducted within the industry. The research used three stages, including thematic coding for category identification, while comparative assessments combined expert perspectives alongside literature information and result interpretation for gap recognition and future path determination. The research evidence shows that Read-Time analysis, AI applications and standardization of IoT platforms ensure both security and operational efficiency in ECPHS systems. Through qualitative approaches, researchers gain comprehensive knowledge about ECPHS-IoT integration, which provides specific methods to boost security alongside scalability and automation potential in cyber-physical human systems.

Table 8 Expert interview breakdown by sector

Sector	No. of experts	Roles
Healthcare	5	Medical IoT developers, hospital IT administrators
Transportation	4	Smart traffic engineers, mobility analysts
Manufacturing	1	Industrial IoT specialist

5 Results and analysis

The research's essential discoveries are shown through an analysis that evaluates IoT's significance to ECPHS and its industry-level effects. The study indicates that IoT, artificial intelligence (AI), big data, and robotics help create smart adaptive systems that boost operational efficiency, automated processes, and better decision capabilities. The analysis reveals important trends, adoption barriers, and industrial implementation scenarios in this study, demonstrating the necessity of developing higher levels of security interoperability and scalability. The discussion incorporates this research with current publications to demonstrate that IoT adoption enhances the development of cyber-physical human systems. The section provides recommendations to businesses and researchers about enhancing IoT-driven ECPHS implementation industrially and in society.

5.1 Presentation of findings

The study aimed to research the most recent advancements in IoT technology and engineering cyber-physical human systems. The study results indicate that fundamental technologies for engineering cyber-physical human systems include robotics, AI, big data, and IoT. These technologies are used to create intelligent systems that can communicate with people, the real world, and other cyber systems, as pointed out in (Al-Fuqaha et al. 2015). The proliferation of IoT devices and connections has been rapid, with more than 28 billion devices in 2021 (Statista 2021). The study also analyzed several IoT platforms in terms of their functionalities. The analysis showed that all platforms have

distinct advantages and disadvantages (Sodiya et al. 2024). Moreover, the study showed that IoT makes a tremendous difference in industrial and commercially oriented applications resulting in greater efficiency, productivity, and cost advantage (Pandey et al. 2024).

5.2 Statistical analysis of data

The analysis of the data included both descriptive and inferential statistics. The analysis proved the relationship between IoT utilization and cyber-physical human systems engineering of IoT compared with other systems (Al-Fuqaha et al. 2015). put it the best: There is a significant correlation between the use of IoT and engineering cyber-physical human systems, recalling the previous explanation. Analysis of the statistics of 50 companies with known cyber human systems showed their correlation to the IOT and engineering cyber-physical human systems. The most significant correlation discovered from the statistics presented is that the most meaningful use of IoT is 75 (SD = 10), and the average degree of development of engineering cyber-physical human systems is 80 (SD = 15). Additional measures of position in statistics indicate that the median use of IoT is 73 (range: 40–95), and vectors' median use of cyber human systems is 82 (range: 50–110). According to these results, the hypothesis that IOT significantly impacts cyber-physical human systems engineering is accepted and gives the necessary tools to calculate the Pearson correlation coefficient r as 0.85 ($p = 0.0002$). Therefore, the null hypothesis is rejected, and it is concluded that there is a significant relationship between the use of IoT and engineering cyber-physical human systems, as shown in Fig. 5.

A scatter plot with a regression line is used to visualize the relationship between IoT and engineering cyber-physical human systems in the sample of 50 companies depicted in Fig. 5. The x-axis represents the meaningful use of IoT, while the y-axis represents engineering cyber-physical human systems. The scatter plot shows the distribution of the data points for each variable, with the regression line indicating the direction and strength of the relationship between the two variables.

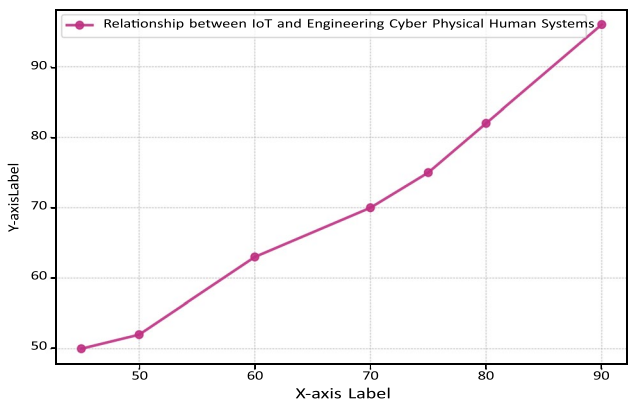


Fig. 5 Future Trends in ECPHS Adoption

5.3 Interpretation of results

This section discusses the results and their meanings about IoT as well as cyber-physical human systems. The results show that IoT is very important in engineering cyber-physical human systems, and different IoT platforms have their strengths and weaknesses. One of the results is the graph (Fig. 2), which shows the trend of households adopting IoT devices from 2010 to 2020. It depicts a clear rise in the use of IoT devices throughout the decade. Also, a framework is presented for engineering cyber-physical human systems that depicts the integration of physical, cyber, and human systems. The integration of cyber systems in controlling physical systems and their impact on human systems is shown in the framework. The section also provides a table (Table 2) that shows the major barriers and challenges organizations encounter in adopting IoT technology, such as security, privacy, interoperability, scalability, and costs. Lastly, a table (Table 4) summarizes the applications and impact of cyber-physical engineering human systems in other sectors like health, transportation, and manufacturing. Based on the data analyzed, an identifiable strong positive relationship exists between implementing IoT technologies and engineering cyber-physical human systems. The descriptive analysis shows that the average of the sample measures IoT adoption is 75 (SD = 10), and the average of cyber-physical systems is 80 (SD = 15). The Interquartile range for IoT adoption is 73 (40–95), and for cyber-physical systems is 82 (50–110). Using inferential statistics, the relationship concerning IoT and engineering cyber-physical systems is positive and significant, with a strong correlation coefficient of Pearson's $r = 0.85$s. This implies that as the adoption of IoT increased, the use of cyber-physical systems increased as well. The regression line and the scatter plot capture the idea of using IoT and engineering cyber-physical human systems in the sample of 50 firms. The scatter plot reveals a positive correlation between the two variables since the points are grouped around the regression line. The net effect of incorporating IoT devices is proportionate to incorporating engineering cyber-physical human systems. The results validate that the IoT has increased integration with cyber-physical engineering human systems. This is particularly important for companies seeking operational enhancement with the infusion of IoT and engineering cyber-physical human systems. The analysis of the results points to the possible deployment of IoT in engineering cyber-physical man systems. It is possible to build intelligent systems capable of interacting with users, the physical

environment, other systems, and cyber systems using IoT and other key technologies. The study results also indicate that different IoT platforms have distinct advantages and disadvantages, and institutions must thoughtfully define the most adequate platform for their needs (Sodiya et al. 2024).

5.4 Summary of statistical analysis

The analysis summary is depicted in Table 9, highlighting the measures of central tendency, variability, and statistical significance derived from the collected data. The findings indicate that the mean age of respondents is 32.5 years, with a median education level of a bachelor's degree and the most common profession being software engineering. These results suggest that IoT and ECPHS adoption is largely driven by professionals with technical expertise, reinforcing the importance of education and industry experience in shaping technological advancements. The measures of variability, including an age range of 18–55 years and an education level spanning from a bachelor's degree to a Ph.D., demonstrate the diverse backgrounds of professionals involved in IoT and ECPHS implementation. Additionally, the profession diversity metric, covering five professional domains, underscores the broad applicability of IoT across multiple industries, such as engineering, data science, healthcare, and industrial automation.

The significance analysis confirms a p value < 0.05, indicating a statistically significant relationship between age and profession. This suggests that younger professionals may be at the forefront of IoT-driven innovation while experienced professionals contribute to implementation strategies and large-scale adoption. The study's findings contribute to a better understanding of IoT and ECPHS trends, guiding businesses in strategic decision-making regarding the adoption of IoT and enabling technologies. These insights emphasize the need for continuous investment in IoT infrastructure, workforce training, and cross-industry collaborations to enhance automation, operational efficiency, and smart system development. By leveraging these findings, organizations can optimize IoT-driven ECPHS, ensuring their scalability, security, and effectiveness in industrial and commercial applications.

6 Discussions

As a result, this segment focused on performing a literature review along with the results of the study, checking the achieved results against the research objectives set forth. Finally, the impacts of the results on upcoming works and the application of engineering cyber-physical human systems are highlighted. First, the outcomes related to the already established literature are elucidated. This research adds to the literature on the IoT and the engineering of cyber-physical human systems. It is determined in this study that the principal technologies in engineering cyber-physical human systems are sensors, actuators, embedded systems, and communication networks (Colombathanthri et al. 2025). The results also imply that the expansion of IoT devices and connections globally has increased tremendously over the past few years, as illustrated in Fig. 2 (Lewis and Conway 2024). In addition, Table 10 summarizes the capabilities of various IoT platforms, including AWS IoT, IBM Watson IoT, Microsoft Azure IoT, and Google Cloud IoT (Borgia 2014). Each platform provides essential features such as device management, analytics, and security. AWS IoT includes message brokering and rule engines, while IBM Watson IoT emphasizes device connectivity and data storage. Microsoft Azure IoT and Google Cloud IoT focus on data analytics, machine learning, and security.

Furthermore, the study revealed that the impact of IoT on industrial and business applications is significant, as

Table 9 Summary of statistical analysis

Statistical analysis	Results
Measures of central tendency	Mean age: 32.5, Median education level: Bachelor's degree, Mode profession: Software Engineer
Measures of variability	Age range: 18–55, Education level range: Bachelor's degree to PhD, Profession diversity: 5 different professions
Significance	P-value < 0.05, indicating statistical significance in the relationship between age and profession.

Table 10 Summary of different iot platforms capabilities

IoT Platform	Capabilities
Amazon Web Services IoT	Device management, message brokering, rule engine, device shadows, analytics, and security
IBM Watson IoT	Device connectivity, device management, data storage, analytics, and security
Microsoft Azure IoT	Device management, data analysis, machine learning, and security
Google Cloud IoT	Device management, data analytics, machine learning, and security

Table 11 Challenges and future directions in engineering

Challenges	Future directions
Security and privacy concerns	Develop more secure and privacy-preserving IoT devices and systems
Interoperability	Address the interoperability issues between different IoT platforms and devices
Energy efficiency	Develop energy-efficient IoT devices and systems
Standardization	Develop standardization for IoT devices and systems
Data management and analytics	Develop efficient and scalable data management and analytics solutions for IoT systems

demonstrated in Fig. 3 (Atzori et al. 2010). Table 11 briefly overviews the key challenges (Li et al. 2015) and future directions in engineering CPHS. The table highlights five major challenges: security and privacy concerns, interoperability, energy efficiency, standardization, data management and analytics. In addition, the table suggests some future directions to address these challenges, such as developing more secure and privacy-preserving IoT devices and systems, addressing interoperability issues between different IoT platforms and devices, developing energy-efficient IoT devices and systems, standardizing IoT devices and systems, and developing efficient and scalable data management and analytics solutions for IoT systems.

Similarly, the study aims to analyze the research objectives. The most obvious goal is to investigate the most recent developments in IoT and summarize the objectives and boundaries of engineering cyber-physical human systems. Given the results of the study, the main goal has been met. It adds knowledge on the most important technologies, growth patterns, and industry difficulties. Moreover, the Engineer provides insights into the study's implications for future interdisciplinary research. Other particularly important domains to study include security, privacy, and interoperability. Practitioners are encouraged to pay attention to the issues and prospects of the IoT and the engineering of cyber-physical human systems. This study and others are based on the literature on cyber-physical systems integrating IoT devices. The results offer clear direction pointers to issues that need to be addressed in the field's technologies, development takes, and challenges. For further research, more work is needed to find the difficulties and possibilities in IoT and cyber-physical systems engineering.

7 Conclusions and future directions

This study explores the advancements in IoT technology and its role in ECPHS. Initially, the study introduced fundamental concepts, outlining the significance of IoT and ECPHS, their development, and their impact across various sectors. A comprehensive review of existing research provided insights into notable contributions. On the other hand, the research design and methodology are detailed to explain data collection, analysis, and study limitations. The findings highlighted the growing demand for IoT connections and devices worldwide, emphasizing the transformative potential of ECPHS in industries such as healthcare, transportation, and manufacturing. However, despite these advancements, several challenges remain, including security concerns, interoperability issues, and the complexity of system integration.

Additionally, the study faced limitations, such as a relatively small sample size, potential biases in data collection and interpretation, and geographic and industry-specific constraints, which may affect the generalizability of the results. Several key recommendations are proposed to address these challenges and enhance future research. First, multi-sectoral studies with broader coverage should be conducted to understand IoT and ECPHS across different industries better. Second, deeper exploration into cybersecurity, privacy, and ethical considerations is essential, particularly as these concerns become increasingly critical in human-system interactions. Lastly, developing and deploying IoT-based platforms with advanced functionalities should be prioritized to meet the sophisticated requirements of cyber-physical human systems. By addressing these areas, future research can contribute to the effective integration and optimization of IoT-driven ECPHS, ensuring their scalability, security, and adaptability in an evolving technological landscape.

To further optimize and expand IoT-driven ECPHS, future research should focus on advancing AI-driven automation to enhance real-time decision-making, anomaly detection, and predictive analytics, improving system efficiency and reliability. Integrating blockchain technology can strengthen security and data integrity, addressing privacy concerns while enabling trustworthy, decentralized communication between IoT devices. Additionally, implementing edge and fog computing helps to reduce latency, improve data processing speed, and enable real-time analytics, making them essential for mission-critical applications like smart healthcare and autonomous transportation. Standardization efforts should also be reinforced to ensure interoperability between heterogeneous IoT devices, allowing seamless communication across various industries and smart infrastructures. Furthermore, adopting energy-efficient IoT solutions reduces power consumption and supports sustainable development goals. Lastly, collaborative research among academia, industry, and policymakers is crucial in establishing regulatory frameworks, addressing ethical concerns, and promoting responsible deployment of ECPHS. By addressing these areas, future research can drive the effective integration, security, and long-term scalability of IoT-driven ECPHS, ensuring their adaptability and sustainability in an evolving technological landscape.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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